ORBITAL MOTION

SIMULATION AND ANALYSIS USING COMPUTATIONAL METHODS

Author: Bhumika

Class 11 Student

Independent Research Project

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Field of Study:

Physics (*Orbital Mechanics & Simulation*)

This project explores the relationship betwœm orbital velocity, altitude, and time period through computational simulation.

Computational Analysis of Satellite Orbital Dynamics with Altitude Variations and Orbital Decay Extension

Author: Bhumika
Class: 11
School: S.L.D.A.V Public school
Year: 2026

ABSTRACT

Satellites play a crucial role in modern communication, navigation, and scientific exploration. This study presents a computational simulation of satellite orbital dynamics using Newtonian gravitational principles. The simulation was implemented using Python and numerical computation techniques to analyze how orbital velocity and orbital period vary with altitude.

The results demonstrate that orbital velocity decreases with increasing altitude, while orbital period increases. These findings are consistent with theoretical predictions derived from gravitational laws.

Additionally, the study extends beyond ideal conditions by introducing atmospheric drag as a factor influencing orbital decay. This highlights real-world limitations of satellite motion and emphasizes the importance of advanced modeling in orbital mechanics.

1. Introduction

Artificial satellites are essential for communication, weather forecasting, navigation, and scientific research. Their motion is governed primarily by gravitational forces described by Newton's law of universal gravitation.

Orbital mechanics studies the motion of objects under gravitational influence. In a circular orbit, gravitational force provides the centripetal force required to keep a satellite in motion.

This study develops a computational simulation to analyze satellite motion and explores how orbital parameters vary with altitude. It also examines real-world limitations such as atmospheric drag.

2. Methodology

The simulation is based on Newtonian mechanics.

Gravitational force: $F = GMm / r^2$

Orbital velocity: $v = \sqrt{(GM / r)}$

Orbital period: $T = 2\pi r / v$

The model was implemented using Python libraries such as NumPy and Matplotlib. Altitudes ranging from 200 km to 2000 km were analyzed.

Assumptions:

Circular orbit

No atmospheric drag (initially)

Uniform Earth mass

3. Results

The simulation provides a three-dimensional representation of satellite motion.

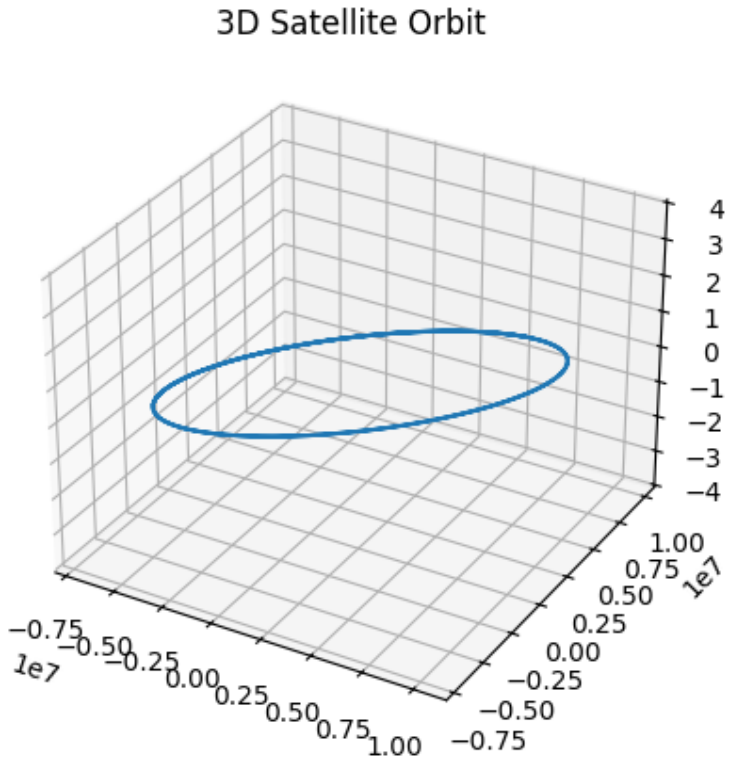

Figure 1: Three-dimensional visualization of a satellite orbit around Earth, illustrating its spatial trajectory under gravitational motion.

A two-dimensional projection provides a clearer orbital view.

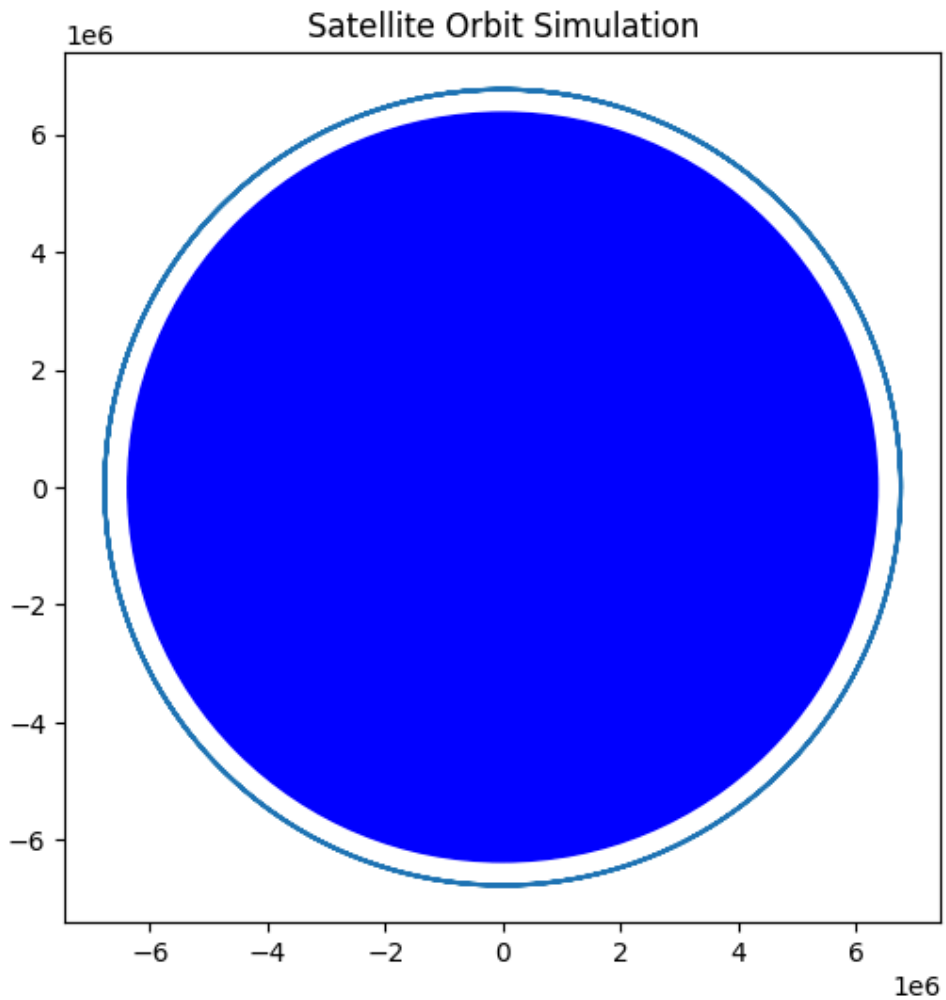

Figure 2: Two-dimensional representation of a satellite orbit around Earth showing a stable circular trajectory.

The relationship between velocity and altitude was analyzed.

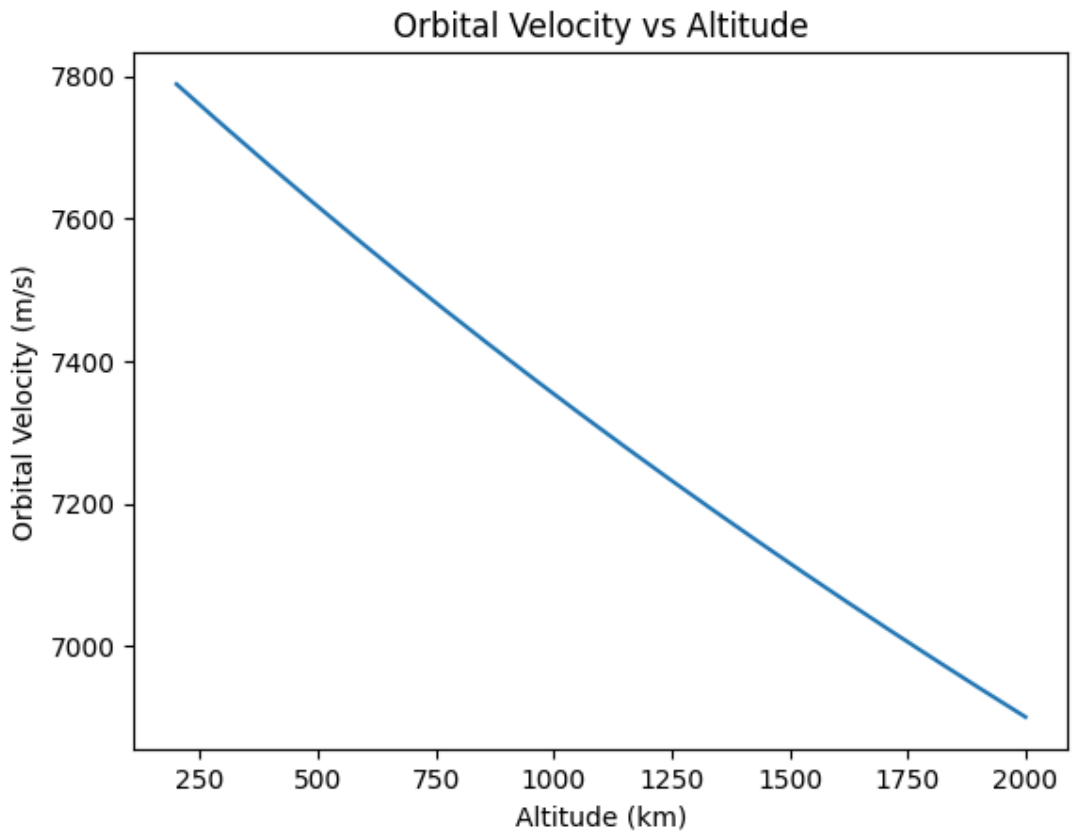

Figure 3: Variation of orbital velocity with altitude, showing a decrease due to reduced gravitational force.

Add: The observed decrease in velocity confirms Newtonian gravitational predictions.

The orbital period variation was analyzed.

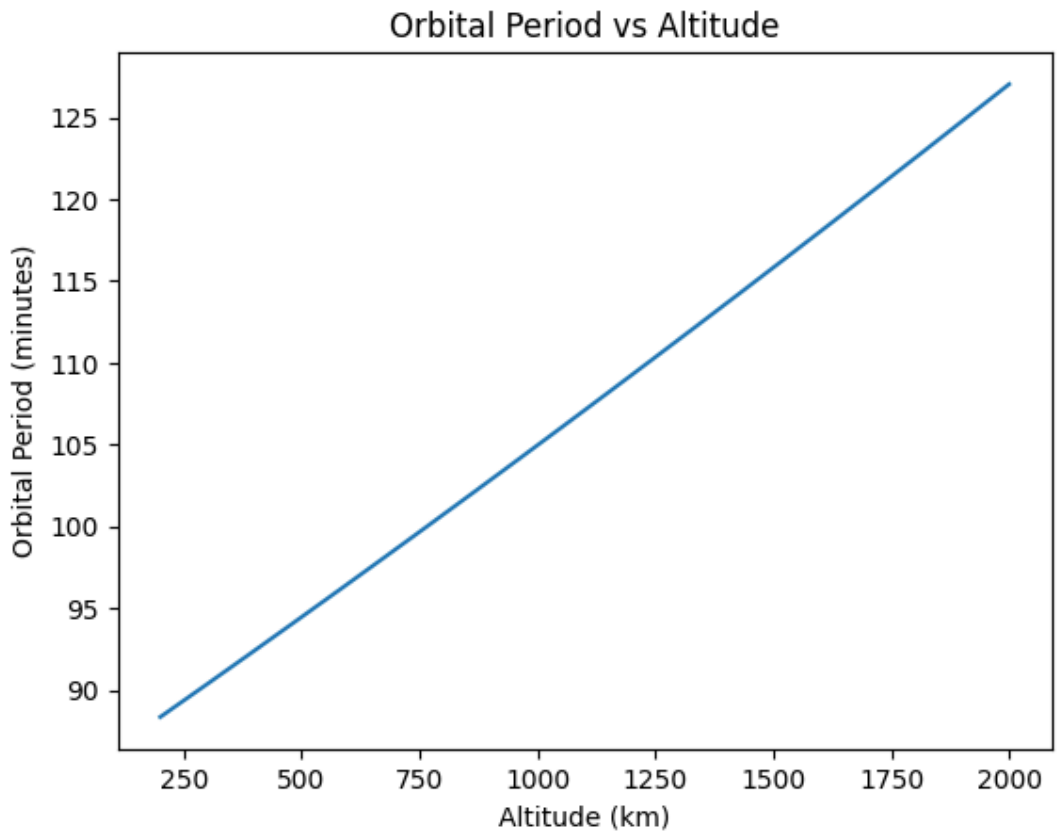

Figure 4: Variation of orbital period with altitude, showing an increase with orbital radius.
Add: This follows Kepler's third law of planetary motion.

A comparative visualization was generated.

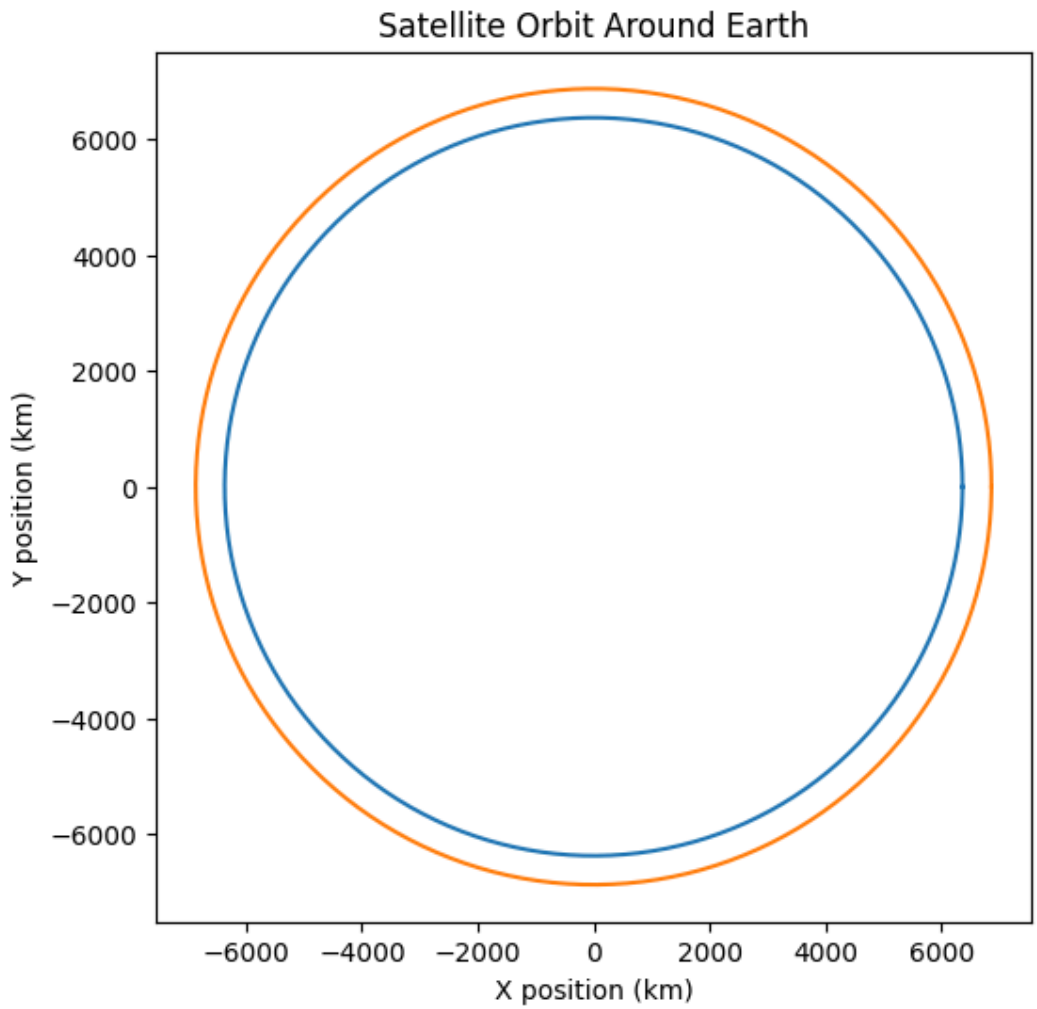

Figure 5: Comparison between Earth's radius and satellite orbital paths at different altitudes.

4. Discussion

The results demonstrate how gravitational forces influence satellite motion. Lower orbits require higher velocity, while higher orbits result in longer periods.
Real-world satellites are affected by atmospheric drag, which leads to orbital decay. This highlights limitations of ideal models and the importance of realistic simulations.

5. Advanced Extension: Orbital Decay

Drag force: $F_d = \frac{1}{2} C_d \rho A v^2$
Atmospheric drag reduces velocity, causing satellites to lose altitude over time.

6. Conclusion

This study successfully models satellite orbital motion using computational methods. The results confirm theoretical relationships between altitude, velocity, and orbital period.

The inclusion of atmospheric drag provides insight into real-world satellite behavior and highlights the importance of advanced modeling techniques.

7. References

Newton, I. (1687). Principia Mathematica

NASA Orbital Mechanics Basics

Fundamentals of Astrodynamics